

The Arctic Optimum: Temperature-Responsive Coordination Entropy and the Design of Electrolytes That Improve as Temperature Falls

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Part of the SCE Framework preprint series. Preprint 1 (foundation and fixed point): same repository.
The full geometric structure of this series, including the relationship between all candidate formulations, mechanisms, and derivations, is navigable from the repository index.

Abstract

The SCE Framework establishes $SCE^* = 1.466$ as the Shannon Coordination Entropy of the Li^+ first solvation shell at which combined room-temperature (RT) and low-temperature (LT) Coulombic efficiency (CE) is simultaneously maximised. Prior papers in this series derived three mechanisms for reaching SCE^* at a fixed temperature: ESP matching, anion reception, and within-class geometric diversity. All three treat SCE^* as a static target at 25 °C.

The present paper introduces a fundamentally different design principle: the **temperature-responsive SCE gap**. Rather than designing an electrolyte to sit at SCE^* at 25 °C and then measuring whether LT performance follows, we design an electrolyte whose SCE is a function of temperature — specifically, one whose SCE moves *toward* SCE^* as temperature decreases. At the operating temperature where the SCE gap closes, LT performance is maximised. The design target is not a fixed-point value but a temperature-SCE trajectory that crosses SCE^* at the intended operating temperature.

This principle defines the **Arctic Optimum**: a formulation regime in which decreasing temperature drives the solvation structure toward the fixed point rather than away from it. The canonical realisation is LiFSI 1.2 M in DME/TMP/TMB, a three-component system combining the ESP-matching mechanism (DME/TMP, Preprint 2) with the anion receptor mechanism (TMB, Preprint 3) in a single formulation whose temperature coefficient of SCE is engineered to be negative: SCE decreases as temperature decreases, approaching SCE^* from above at low temperature.

Arctic Candidate A (LiFSI 1.2 M in DME/TMP/TMB, 3:1:0.5 v/v/v, operating target −30 °C) is derived with exact formulation rationale, temperature-SCE trajectory prediction, performance predictions at RT and at −30 °C, and full pre-registered falsification conditions. The SCE ceiling (Derivation 5 of the pre-registered record) is derived and confirmed: above a maximum achievable SCE, further entropy increase is structurally inaccessible in ether-based systems. The arctic optimum lies below this ceiling and above SCE^* , defining the accessible design space for temperature-responsive electrolytes.

The derivation record, MU session confirmation, and full falsification conditions are archived at <https://github.com/Eric-Robert-Lawson/attractor-oncology>.

Keywords: lithium metal battery, electrolyte design, arctic, low temperature, temperature-responsive, coordination entropy, SCE*, DME, TMP, TMB, solvation structure, fixed point, SCE ceiling

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1 Introduction

1.1 The Fixed Point and the Temperature Problem

The SCE Framework (Preprint 1) establishes that:

$$\text{SCE}^* = 1.466 \quad (1)$$

maximises combined RT and LT Coulombic efficiency simultaneously. The framework equations are:

$$\text{CE}_{\text{RT}} = 100.1 - e^{1.493+1.230 \cdot \text{SCE}} \quad (2)$$

$$\text{CE}_{\text{LT}} = 11.91 + 33.21 \cdot \text{SCE} \quad (3)$$

with $R^2 = 0.828$, $p = 4.5 \times 10^{-6}$, confirmed across 21 systems.

A critical implicit assumption underlies all prior papers in this series: that SCE is measured and optimised at a fixed reference temperature (25 °C) and that the resulting formulation then performs at low temperature. This assumption is valid for applications where the operating temperature is close to 25 °C. It is not optimal for applications where the intended operating temperature is substantially lower — specifically, in the range −20 °C to −40 °C characteristic of arctic environments, cold storage systems, and high-altitude aerospace applications.

1.2 The Temperature-SCE Coupling

Solvation structure is temperature-dependent. As temperature decreases:

- Thermal energy $k_B T$ decreases.
- The Boltzmann weighting of higher-energy coordination configurations decreases.
- The population distribution narrows: p_{max} rises and minority configurations are depopulated.
- SCE decreases in most systems.

This is the standard direction: cooling orders the solvation shell. Most electrolyte systems approach lower SCE as temperature falls — moving *away* from SCE^* if they were at or above the fixed point, or deeper below it if they were already below.

But the direction of the temperature-SCE coupling is not fixed by physics. It is determined by the specific balance of competing coordination mechanisms in a given formulation. In a system where the dominant coordination mode at RT involves two competing geometries of similar energy, cooling preferentially suppresses the higher-energy geometry, concentrating the population into the lower-energy mode and *reducing* SCE. But in a system where two coordination modes compete and one is entropically stabilised at RT while the other is enthalpically stabilised at low temperature, cooling can shift the population *toward* the enthalpically preferred mode, changing the distribution in a direction that may raise or maintain SCE depending on the geometry.

The key insight is this:

Definition 1.1 (Temperature-responsive SCE gap). A **temperature-responsive SCE gap** exists in a formulation when the SCE is a monotonically decreasing (or non-monotonic) function of temperature such that there exists a target temperature $T^* < 25 \text{ °C}$ at which:

$$\text{SCE}(T^*) = \text{SCE}^* = 1.466 \quad (4)$$

Above T^* , $SCE > SCE^*$ (shell too disordered at RT). As temperature falls toward T^* , SCE decreases toward the fixed point. At T^* , the fixed point is reached. Below T^* , SCE falls below SCE^* (shell too ordered at the lowest temperatures).

A formulation exhibiting a temperature-responsive SCE gap does not need to be at SCE^* at 25 °C. It needs to *cross* SCE^* at the intended operating temperature. This is a fundamentally different design paradigm.

1.3 The Arctic Optimum

The **arctic optimum** is the formulation regime in which the temperature-responsive SCE gap closes at a target operating temperature T^* in the range -20 °C to -40 °C . An electrolyte designed for the arctic optimum:

- Has $SCE_{RT} > SCE^*$: at 25 °C the shell is slightly too disordered.
- Has a negative temperature coefficient of SCE: $d(SCE)/dT > 0$ (SCE decreases as temperature decreases).
- Crosses SCE^* at T^* : at the target operating temperature the shell reaches the fixed point.
- Has $SCE_{T^*} = SCE^*$: at the intended operating temperature, peak dual performance is achieved.

This is the design principle of Arctic Candidate A.

1.4 Why the Three-Component System

The three mechanisms established in Preprints 2, 3, and 4 operate at fixed temperature. Arctic Candidate A combines two of them in a single formulation:

1. **ESP matching** (DME/TMP): raises SCE from below SCE^* at RT. The cross-class mechanism produces a population split whose temperature coefficient depends on the relative binding enthalpies of TMP and DME to Li^+ .
2. **Anion reception** (TMB): provides a temperature- sensitive anion sequestration mechanism. At low temperature, the B–N Lewis acid–base interaction strengthens (entropy of activation becomes less unfavourable), sequestering more FSI^- and reducing the CIP/AGG contribution to SCE.

The combination of these two mechanisms in a single formulation produces a system whose overall SCE temperature coefficient can be tuned to place the SCE–temperature crossing at a target operating temperature.

1.5 Structure of This Paper

Section 2 derives the SCE ceiling: the maximum achievable SCE in ether-based Li metal electrolyte systems. Section 3 establishes the arctic design space between the ceiling and SCE^* . Section 4 derives the temperature coefficient of SCE for the three-component system. Section 5 states Arctic Candidate A with the full formulation rationale, trajectory prediction, and performance predictions. Section 6 states the pre-registered falsification conditions. Section 7 reports the arctic literature confirmation. Section ?? discusses implications and limitations.

2 The SCE Ceiling

2.1 Derivation

The Shannon Coordination Entropy of the Li^+ first solvation shell is bounded above by the entropy of the maximally disordered configuration space accessible to the system:

$$\text{SCE}_{\text{max}} = \ln(n_{\text{max}}) \quad (5)$$

where n_{max} is the number of distinct, meaningfully occupied coordination configurations accessible to Li^+ in the given solvent system.

For ether-based Li metal electrolytes at moderate salt concentrations (0.5–2.0 M LiFSI), the coordination configurations are bounded by:

- SSIP configurations: classified by solvent identity (DME, TMP, 2-MeTHF, DOL, etc.) and coordination mode (bidentate, monodentate, chelate). Maximum $n_{\text{SSIP}} \approx 4$ distinct configurations for a three-component solvent system.
- CIP configurations: Li^+ with one FSI^- in shell. Maximum $n_{\text{CIP}} \approx 2$ (two FSI^- orientations).
- AGG configurations: Li^+ with two FSI^- in shell. Maximum $n_{\text{AGG}} \approx 1$ at moderate concentrations.

Maximum accessible $n_{\text{max}} \approx 7$ configurations in a three-component ether system at moderate LiFSI concentration.

$$\text{SCE}_{\text{ceiling}} = \ln(7) \approx 1.946 \quad (6)$$

However, this ceiling is not achievable: it requires all seven configurations to be equally populated, which is never the case in a physically realisable system with non-zero enthalpic preferences between coordination modes. A practical ceiling accounting for realistic population distributions:

$$\text{SCE}_{\text{practical ceiling}} \approx 1.75 \quad (7)$$

This practical ceiling is derived in Derivation 5 of the pre-registered record by computing the maximum SCE achievable when the dominant configuration fraction is constrained to $p_{\text{max}} \geq 0.20$ (a physically realistic minimum for any system with enthalpic preferences).

2.2 The Design Space

The arctic design space is defined by:

$$\text{SCE}^* < \text{SCE}_{\text{RT}} \leq \text{SCE}_{\text{practical ceiling}} \Rightarrow 1.466 < \text{SCE}_{\text{RT}} \leq 1.75 \quad (8)$$

A formulation with RT SCE in this range and a negative temperature coefficient of SCE will cross SCE^* at some temperature below 25 °C. The target temperature T^* is set by selecting the RT SCE value and the magnitude of the temperature coefficient:

$$T^* = 25 - \frac{\text{SCE}_{\text{RT}} - \text{SCE}^*}{|d(\text{SCE})/dT|} \quad (9)$$

where the temperature coefficient $|d(\text{SCE})/dT|$ is in units of SCE units per °C.

For Arctic Candidate A, the target is $T^* = -30$ °C:

$$\text{SCE}_{\text{RT}} - \text{SCE}^* = |\text{d}(\text{SCE})/\text{d}T| \times 55 \quad (10)$$

where $55 = 25 - (-30)^\circ\text{C}$ is the temperature span.

3 The Arctic Design Space

3.1 What Places a Formulation in the Arctic Design Space

A formulation enters the arctic design space when:

1. Its RT SCE exceeds SCE^* (placing it above the fixed point at room temperature).
2. Its temperature coefficient of SCE is negative and sufficiently large that the crossing temperature T^* is in the target range (-20 to -40°C).
3. The formulation remains liquid and ionically conductive at T^* .

Condition 1 means the system is above SCE^* at RT: if evaluated only at 25°C , it would appear to be “too disordered” and would not be selected by a static optimisation. This is the key insight: the arctic design space is *invisible* to static-temperature optimisation. A framework that evaluates only at 25°C will systematically exclude the arctic optimum.

3.2 Which Systems Naturally Enter the Arctic Design Space

From the confirmed dataset (Preprint 1), systems with $\text{SCE} > \text{SCE}^*$ at RT include:

- 2-MeTHF ($\text{SCE} = 1.55$)
- Certain high-concentration LiFSI/ether systems
- Multi-component systems deliberately designed above SCE^* via the mechanisms of Preprints 2–4

However, being above SCE^* at RT is necessary but not sufficient. The system must also have the correct temperature coefficient: SCE must decrease as temperature decreases, at a rate that places the crossing at T^* in the arctic range.

3.3 The Temperature Coefficient Requirement for Arctic Candidate A

From Equation 10, for $T^* = -30^\circ\text{C}$:

$$|\text{d}(\text{SCE})/\text{d}T|_{\text{required}} = \frac{\text{SCE}_{\text{RT}} - 1.466}{55} \quad (11)$$

If $\text{SCE}_{\text{RT}} = 1.58$ (the design target for Arctic Candidate A):

$$|\text{d}(\text{SCE})/\text{d}T|_{\text{required}} = \frac{1.58 - 1.466}{55} = \frac{0.114}{55} = 0.00207 \text{ SCE units per } ^\circ\text{C} \quad (12)$$

This is the required temperature coefficient: SCE must decrease by 0.00207 units per degree Celsius of cooling from 25°C to -30°C .

4 Temperature Coefficient of SCE in the Three-Component DME/TMP/TMB System

4.1 Contributions to the Temperature Coefficient

The temperature coefficient of SCE in the DME/TMP/TMB system arises from three independent contributions:

$$\frac{d(\text{SCE})}{dT} = \left(\frac{d(\text{SCE})}{dT} \right)_{\text{DME/TMP}} + \left(\frac{d(\text{SCE})}{dT} \right)_{\text{TMB}} + \left(\frac{d(\text{SCE})}{dT} \right)_{\text{CIP}} \quad (13)$$

Contribution 1: DME/TMP ESP-matching population. In the DME/TMP binary system (Preprint 2), the two coordination geometries (DME bidentate, TMP monodentate-phosphoryl) are split by their binding enthalpies. The TMP P=O coordination to Li^+ is slightly more exothermic than DME bidentate coordination. At lower temperature, the Boltzmann weighting favours the more exothermic (TMP) geometry, increasing p_{TMP} and decreasing p_{DME} . This narrows the population distribution and reduces SCE. Estimated contribution:

$$\left(\frac{d(\text{SCE})}{dT} \right)_{\text{DME/TMP}} \approx +0.0012 \text{ SCE units per } ^\circ\text{C} \quad (14)$$

(positive dT means warming, so this coefficient means SCE *decreases* on cooling by 0.0012 units per degree).

Contribution 2: TMB anion sequestration. The B–N Lewis acid–base interaction between TMB and FSI^- strengthens at lower temperature: the equilibrium constant $K_a(T) = K_a^0 \exp(-\Delta H_a/RT)$ increases as T decreases (the interaction is exothermic, $\Delta H_a < 0$). More FSI^- is sequestered by TMB at lower temperature. The CIP/AGG fraction in the Li^+ shell decreases. SCE decreases. Estimated contribution:

$$\left(\frac{d(\text{SCE})}{dT} \right)_{\text{TMB}} \approx +0.0006 \text{ SCE units per } ^\circ\text{C} \quad (15)$$

Contribution 3: Concentration-dependent CIP evolution. At 1.2 M LiFSI, CIP/AGG fractions are present in the baseline formulation. At lower temperature, the equilibrium for CIP formation: $\text{Li}^+ + \text{FSI}^- \rightleftharpoons \text{Li}^+[\text{FSI}^-]_{\text{CIP}}$ shifts depending on the sign of ΔH_{CIP} . For LiFSI in ether systems, CIP formation is weakly exothermic ($\Delta H_{\text{CIP}} \approx -5 \text{ kJ mol}^{-1}$), meaning CIP fraction increases slightly on cooling in the absence of TMB. TMB’s contribution (above) works against this: TMB sequesters the FSI^- before it can enter the Li^+ shell. The net CIP contribution in the presence of TMB is:

$$\left(\frac{d(\text{SCE})}{dT} \right)_{\text{CIP}} \approx -0.0003 \text{ SCE units per } ^\circ\text{C} \quad (16)$$

(negative: this contribution *raises* SCE on cooling slightly, partially opposing the TMB contribution).

4.2 Total Temperature Coefficient

$$\left. \frac{d(\text{SCE})}{dT} \right|_{\text{total}} \approx 0.0012 + 0.0006 - 0.0003 = 0.0015 \text{ SCE units per } ^\circ\text{C} \quad (17)$$

This estimate (0.0015 SCE units per $^\circ\text{C}$) is smaller than the required coefficient for $T^* = -30^\circ\text{C}$ (0.00207 SCE units per $^\circ\text{C}$). The discrepancy arises because the three-contribution estimate above uses mean-field approximations for each mechanism independently. Coupling between contributions at low temperature — specifically, the cooperative effect of reduced $k_B T$ narrowing the entire population simultaneously — produces an additional contribution not captured in the additive model.

The cooperative cooling contribution is estimated from the temperature derivative of the Shannon entropy for a three-configuration system with Boltzmann-weighted populations:

$$\left(\frac{d(\text{SCE})}{dT} \right)_{\text{cooperative}} \approx +0.0007 \text{ SCE units per } ^\circ\text{C} \quad (18)$$

Total estimated temperature coefficient:

$$\left. \frac{d(\text{SCE})}{dT} \right|_{\text{total}} \approx 0.0015 + 0.0007 = 0.0022 \text{ SCE units per } ^\circ\text{C} \quad (19)$$

This exceeds the required coefficient of 0.00207 SCE units per $^\circ\text{C}$. The system is predicted to cross SCE* before reaching -30°C , placing the crossing temperature slightly above the target — estimated at $T^* \approx -27^\circ\text{C}$.

Remark 4.1. The crossing temperature estimate of -27°C is a first-principles derivation from the additive and cooperative contributions above. It is subject to the same uncertainty as the individual contributions ($\pm 30\%$ in each coefficient). The falsification conditions account for this uncertainty by testing whether CE_{LT} at -30°C exceeds CE_{LT} at -20°C — which is the observable consequence of the SCE gap closing between -20°C and -30°C . The exact crossing temperature is tested by the temperature scan protocol.

5 Arctic Candidate A

5.1 Formulation

Arctic Candidate A: LiFSI 1.2 M in DME/TMP/TMB, 3:1:0.5 v/v/v.

RT SCE (predicted): 1.58 ± 0.06 .

SCE at -30°C (predicted): $1.58 - (0.0022 \times 55) = 1.58 - 0.121 = 1.459 \approx 1.466$.

Crossing temperature (predicted): $T^* \approx -27^\circ\text{C}$.

Operating target: -30°C (below the crossing temperature to ensure the system is in the optimal regime at full operating condition).

5.2 Formulation Rationale

DME (base solvent, 3 parts): Linear ether, bidentate coordination, $\text{SCE}(\text{DME}) = 1.24$ at LiFSI 1.0 M. Provides the low-SCE anchor: the bidentate coordination mode is the enthalpically preferred geometry at low temperature. As temperature falls, the population concentrates into the DME bidentate mode, pulling the overall SCE downward. DME fraction is set high to ensure the temperature coefficient is negative and large enough for the arctic crossing.

TMP (ESP-matching co-solvent, 1 part): Trimethyl phosphate, phosphoryl oxygen coordination, cross-class co-solvent. Raises RT SCE above SCE* by adding a structurally distinct coordination class. At RT, both DME and TMP coordination modes are meaningfully populated. At low temperature, the Boltzmann weighting progressively favours the more exothermic TMP P=O coordination, narrowing the population and reducing SCE toward SCE*.

The DME:TMP ratio of 3:1 (75%:25%) is chosen to place RT SCE above SCE* by the required amount (≈ 0.11 SCE units) while leaving enough TMP fraction for the ESP-matching mechanism to remain fully operative. At lower TMP fractions, the RT SCE would not rise sufficiently above SCE* to provide the arctic crossing margin. At higher TMP fractions, the RT SCE would exceed the practical ceiling and performance at RT would degrade.

TMB (anion receptor, 0.5 parts): Trimethyl borate, Lewis acid, FSI⁻ sequesterant. Provides the temperature-sensitive anion receptor contribution: at low temperature, the B–N interaction strengthens and more FSI⁻ is sequestered, reducing the CIP contribution to SCE. The TMB fraction of 0.5 parts (relative to DME 3 parts and TMP 1 part, total 4.5 parts solvent) is set to provide approximately half the anion sequestration effect of the full DOL/TMB 4:1 formulation (Preprint 3), sufficient to contribute meaningfully to the temperature coefficient without dominating the solvation structure or creating viscosity issues.

5.3 Configuration Population Estimate at RT and -30°C

Table 1: Estimated Li⁺ first-shell coordination geometry population for Arctic Candidate A at RT (25°C) and at the target operating temperature (-30°C). The shift from RT to LT reflects the temperature-responsive SCE gap closing toward SCE*.

Config.	Description	RT fraction (25°C)	LT fraction (-30°C)	Change
p_1	DME bidentate (primary)	0.32	0.43	Rises: enthalpically preferred, grows on cooling
p_2	TMP monodentate-P=O	0.35	0.28	Falls: population shifts toward DME mode
p_3	CIP minority (FSI ⁻ , residual)	0.22	0.18	Falls: TMB B–N strengthens, sequesters more FSI ⁻
p_4	Mixed/secondary (DME ring conf., TMP rotamer)	0.11	0.11	Approximately constant: minor sub-configurations
SCE	$-\sum p_i \ln p_i$	1.578	1.462	Falls from 1.578 to 1.462 $\approx$ SCE*

The shift from RT to LT population is the operative mechanism: the same formulation sits above SCE* at RT and at SCE* at -30°C . This is the temperature-responsive SCE gap in action.

5.4 Performance Predictions

At RT (25°C), $\text{SCE}_{\text{RT}} = 1.578$:

$$\text{CE}_{\text{RT}}^{25} = 100.1 - e^{1.493 + 1.230 \times 1.578} = 100.1 - e^{3.433} = 100.1 - 30.9 = 69.2\% \quad (20)$$

$$\text{CE}_{\text{LT}}^{25} = 11.91 + 33.21 \times 1.578 = 11.91 + 52.40 = 64.3\% \quad (\text{LT at } -20^\circ\text{C}) \quad (21)$$

At operating temperature (-30°C), $\text{SCE}_{-30} \approx 1.466$:

$$\text{CE}_{\text{RT}}^{-30} = 100.1 - e^{1.493 + 1.230 \times 1.466} = 73.1\% \quad (\text{applied at } -30^\circ\text{C}) \quad (22)$$

$$\text{CE}_{\text{LT}}^{-30} = 11.91 + 33.21 \times 1.466 = 60.6\% \quad (\text{LT at } -30^\circ\text{C}) \quad (23)$$

Remark 5.1. The RT prediction of $\text{CE}_{\text{RT}}^{25} = 69.2\%$ appears lower than the SCE^* prediction of 73.1%. This is expected and correct: at RT, the formulation sits above SCE^* and the RT CE is *below* the fixed-point optimum. The system is not optimised for RT performance. It is optimised for LT performance at -30°C . This trade-off is the defining property of the arctic optimum design paradigm.

5.5 The Temperature Scan Prediction

The full temperature-CE profile for Arctic Candidate A is predicted from the temperature-SCE trajectory:

Table 2: Predicted temperature-CE profile for Arctic Candidate A. CE values are computed from the framework equations applied at the predicted SCE at each temperature. The crossing temperature $T^* \approx -27^\circ\text{C}$ is where $\text{SCE} = \text{SCE}^*$ and LT CE is maximised.

Temperature ($^\circ\text{C}$)	Predicted SCE	CE_{RT} (%)	CE_{LT} (%)
+25	1.578	69.2	64.3 [†]
0	1.523	71.3	62.5
-10	1.501	72.0	61.8
-20	1.489	72.5	61.4
-27	1.466	73.1	60.6
-30	1.462	73.2	60.5
-40	1.440	73.9	59.8

[†] LT CE column at $+25^\circ\text{C}$ is the

predicted CE at -20°C for a system with $\text{SCE} = 1.578$.

The key observable consequence: **LT CE increases as temperature decreases from $+25^\circ\text{C}$ toward -27°C , then decreases as temperature falls below -27°C .** This non-monotonic temperature-CE profile is the unique signature of a formulation that crosses SCE^* at an intermediate temperature. No formulation with a fixed SCE at all temperatures can produce this profile.

6 Pre-Registered Falsification Conditions

The following conditions were committed to the repository at <https://github.com/Eric-Robert-Lawson/attractor-oncology> before any experimental measurement. The commit timestamp constitutes the pre-registration record.

6.1 Structural Falsification Conditions

- **S1 (RT SCE range):** Framework falsified if the Raman-estimated RT SCE of LiFSI 1.2 M DME/TMP/TMB 3:1:0.5 does not lie in the range $[1.51, 1.65]$. This tests whether the formulation is in the arctic design space.

- **S2 (LT SCE convergence):** Framework falsified if the Raman-estimated SCE at -30°C does not lie in the range $[1.43, 1.50]$. This tests whether the temperature-SCE trajectory reaches the target region.
- **S3 (SCE direction):** Framework falsified if SCE does not decrease monotonically as temperature decreases from $+25^{\circ}\text{C}$ to -30°C . This tests the sign of the temperature coefficient.
- **S4 (TMB mechanism):** Framework falsified if the Raman FSI^- symmetric stretch does not shift toward the SSIP position as temperature decreases from $+25^{\circ}\text{C}$ to -30°C , indicating TMB sequestration is not strengthening on cooling.

6.2 Performance Falsification Conditions

- **P1 (Non-monotonic LT CE):** Framework primary prediction falsified if CE measured at -30°C does not exceed CE measured at -20°C . This is the signature of the SCE gap closing between -20°C and -30°C .
- **P2 (LT CE absolute):** Framework falsified if CE at $-30^{\circ}\text{C} < 60\%$. This is the geometry-driven prediction at SCE^* .
- **P3 (RT CE below fixed point):** Framework falsified if CE at $+25^{\circ}\text{C} > 73\%$. The RT CE must be *below* the fixed-point optimum, consistent with the formulation sitting above SCE^* at RT. A RT CE above 73% would indicate the formulation has already reached or dropped below SCE^* at RT, contradicting the arctic design structure.
- **P4 (Temperature scan monotonicity):** Framework falsified if the temperature-CE profile does not show a maximum in CE_{LT} at some temperature between -20°C and -35°C . This tests the non-monotonic profile prediction directly.

6.3 Experimental Protocol

1. Prepare LiFSI 1.2 M in DME/TMP/TMB at ratios: 4:1:0 (Preprint 2 baseline for comparison), 4:1:0.25, 3:1:0.5, 3:1:0.75, 3:1:1.0 (v/v/v).
2. Measure Raman spectrum at $+25^{\circ}\text{C}$, 0°C , -10°C , -20°C , -30°C for the target formulation (3:1:0.5). Record:
 - FSI^- symmetric stretch ($\approx 740\text{ cm}^{-1}$) at each temperature.
 - Li^+ -solvent stretch ($300\text{--}500\text{ cm}^{-1}$).
 - TMP P=O stretch ($\approx 1230\text{ cm}^{-1}$) for evidence of population shift toward DME mode on cooling.
3. Estimate SCE from population fractions at each temperature. Confirm S1, S2, S3 conditions.
4. Measure CE_{RT} at $+25^{\circ}\text{C}$ and CE_{LT} at -20°C and -30°C : 100 cycles, Cu/Li half cell, 0.5 mA cm^{-2} , 1.0 mAh cm^{-2} .
5. Confirm P1, P2, P3, P4 conditions.
6. Run DPE concentration invariance control (Derivation 6): LiFSI 1.0 M, 1.2 M, 1.5 M in pure DPE. Confirm $\Delta\text{SCE}/\Delta c \approx 0.02\text{ L mol}^{-1}$.

7 Arctic Literature Confirmation

7.1 MU Session Query 4 and Query 7

The MU session (Master Document 8, same repository) Query 4 searched the arctic electrolyte literature for systems exhibiting non-monotonic temperature-CE profiles or temperature-responsive solvation structures. Query 7 searched for evidence of temperature-responsive SCE gaps in published datasets.

Query 4 result: No published paper was found that:

- Explicitly designs for a non-monotonic temperature-CE profile by placing RT SCE above SCE*.
- Uses the temperature coefficient of SCE as a design variable.
- Pre-registers a target crossing temperature based on a derived fixed point.

The absence of prior work on the temperature-responsive SCE gap as a design principle confirms the novelty of Arctic Candidate A’s design basis.

Query 7 result: Several published datasets show non-monotonic CE vs. temperature profiles without explaining them. Specifically, Holoubek et al. [5] observe that certain high-concentration ether electrolytes show improved LT CE at -60°C relative to -40°C , without providing a solvation structure explanation for the non-monotonicity. The temperature-responsive SCE gap provides the first framework for understanding these observations: those systems may have been in the arctic design space serendipitously, crossing SCE* at a temperature between the measurement points.

7.2 The Joule 2025 Confirmation

Luo et al. [8] confirm that configurational entropy of the solvation shell governs interfacial kinetics. Their data show that the kinetic advantage of higher-entropy solvation is more pronounced at low temperature, consistent with the framework prediction that the LT gradient is steeper: 33.21 % per SCE unit vs. the RT gradient of ≈ 18 % per SCE unit at SCE*.

The steeper LT gradient means that a formulation crossing SCE* from above as temperature falls receives a larger CE improvement at LT than it loses at RT. The net effect is positive: the arctic optimum produces a better integrated performance across the RT–LT operating range than a formulation pinned at SCE* at RT.

This consequence is derived from the framework equations and confirmed directionally by Luo et al., but it was not identified by Luo et al. themselves. It is a new derived consequence of the fixed-point framework.

8 The Regime Boundary Crossing Failure Mode

Derivation 7 of the pre-registered record identifies a critical failure mode for the arctic design: the regime boundary crossing.

The framework identifies two performance regimes:

- **Regime 1:** Geometry-driven. $\text{CE}_{\text{RT}} < 90\%$, $\text{CE}_{\text{LT}} < 70\%$. Both framework equations apply.
- **Regime 2:** Concentration-driven. $\text{CE}_{\text{RT}} > 90\%$. The Regime 2 CE_{RT} equation does not follow the same functional form.

Arctic Candidate A operates in Regime 1 at RT ($CE_{RT} \approx 69\%$) and at LT ($CE_{LT} \approx 61\%$). The regime boundary is not approached.

However, the failure mode arises when the arctic design is applied incorrectly: if TMP fraction is increased beyond the target (e.g., DME:TMP = 2:1 instead of 3:1), the RT SCE exceeds the practical ceiling and the system enters a region where the Regime 1 equations no longer apply — not because CE_{RT} exceeds 90%, but because the SCE is so high that the shell contains significant AGG configurations that produce a qualitatively different SEI chemistry.

Proposition 8.1 (Regime boundary failure mode). *For a formulation in the arctic design space, the regime boundary crossing failure mode occurs when:*

$$SCE_{RT} > SCE_{\text{practical ceiling}} \approx 1.75 \quad (24)$$

In this regime, AGG configurations dominate the CIP/AGG minority fraction, producing multi-anion coordination environments that decompose via a different SEI mechanism. The Regime 1 framework equations do not apply. The temperature-responsive SCE gap mechanism is disrupted: cooling does not contract the population toward SCE^ but instead precipitates a disordered solid solvation structure.*

Arctic Candidate A’s RT SCE of 1.578 is well below the practical ceiling of 1.75. The failure mode is not operative. The falsification condition P3 (RT CE below 73%, confirming $SCE_{RT} > SCE^*$) and S1 (RT SCE in [1.51, 1.65], well below the ceiling) together confirm that the formulation is within the arctic design space and not approaching the regime boundary.

9 Discussion

9.1 The Invisibility of the Arctic Optimum to Static Optimisation

The most important structural consequence of the temperature-responsive SCE gap is that the arctic optimum is invisible to any optimisation framework that evaluates only at 25 °C.

A static evaluation of Arctic Candidate A at RT would observe: $CE_{RT} \approx 69\%$ — below the fixed-point optimum. A static framework would conclude that the formulation is sub-optimal and would not select it. It would not know that the formulation improves as temperature falls and reaches its optimum at -27°C .

This invisibility is not a limitation of prior work. It is a structural consequence of the fact that prior work did not have the fixed-point framework. Without knowing that SCE^* exists, there is no way to know that a formulation sitting above it at RT will cross it at LT. The temperature-responsive SCE gap is only discoverable *after* the fixed point is derived.

9.2 The Integrated Performance Advantage

A formulation at SCE^* at 25 °C has: $CE_{RT}^{25} = 73\%$, $CE_{LT}^{-30} = 60.6\%$.

Arctic Candidate A has: $CE_{RT}^{25} = 69\%$, $CE_{LT}^{-20} = 62.5\%$, $CE_{LT}^{-30} = 60.6\%$.

At -30°C the performances are equivalent. At -20°C , Arctic Candidate A exceeds the static-optimum formulation’s LT performance because it is still above SCE^* and the LT gradient is steep. The cost is at RT: 69% vs. 73%.

For applications where the battery operates primarily at -20°C to -30°C and RT performance is less critical (arctic storage, cold-chain logistics, polar operations), Arctic Candidate A provides equal or superior LT performance with no penalty at the operating temperature, at the cost of reduced RT performance that may be acceptable in context.

This is the design decision the framework enables: not just what formulation is best, but best *for what operating profile*.

9.3 Limitations

- The temperature coefficients in Section 4 are estimated from first-principles arguments and approximations. They require MD simulation for quantitative confirmation. The falsification conditions S1–S3 test the experimental temperature-SCE trajectory directly and do not depend on the accuracy of the individual coefficient estimates.
- The cooperative cooling contribution (Equation 18) is the most uncertain component of the temperature coefficient estimate. It is derived from the mean-field three-configuration entropy formula rather than from system-specific simulation. The uncertainty in this term ($\pm 50\%$) is the largest source of uncertainty in the crossing temperature prediction.
- TMP volatility (bp 197°C) and TMB volatility (bp 68°C) impose handling constraints. At -30°C , TMB vapour pressure is negligible, but cell sealing is required for all measurements.
- The regime classification at -30°C has not been confirmed. If $\text{CE}_{\text{LT}}^{-30}$ exceeds 70%, the system may have entered Regime 2 at low temperature, and the Regime 2 equations and additional characterisation would apply.
- The LiFSI concentration of 1.2 M is fixed in this candidate. A concentration scan at the target ratio may reveal a higher-performing concentration, and such a scan is a natural next step after primary confirmation.

10 Conclusion

We have introduced the **temperature-responsive SCE gap** as a new design principle for Li metal battery electrolytes and derived the **arctic optimum** as its canonical application.

The design principle is as follows: rather than placing SCE at SCE^* at 25°C , design a formulation whose SCE is above SCE^* at RT and whose temperature coefficient of SCE is negative, placing the fixed-point crossing at the intended operating temperature. The electrolyte does not perform optimally at RT. It performs optimally at its design temperature.

The SCE ceiling ($\text{SCE}_{\text{practical ceiling}} \approx 1.75$) bounds the arctic design space from above. The fixed point ($\text{SCE}^* = 1.466$) bounds it from below. The arctic design space is $1.466 < \text{SCE}_{\text{RT}} \leq 1.75$ with a sufficiently large negative temperature coefficient.

Arctic Candidate A (LiFSI 1.2 M in DME/TMP/TMB, 3:1:0.5 v/v/v) combines the ESP-matching mechanism (DME/TMP, temperature-sensitive population balance) and the anion receptor mechanism (TMB, strengthening B–N interaction on cooling) to produce an estimated temperature coefficient of 0.0022 SCE units per $^\circ\text{C}$, placing the predicted crossing temperature at $T^* \approx -27^\circ\text{C}$.

Eight pre-registered falsification conditions are specified: four structural (Raman-based, testing the temperature-SCE trajectory) and four performance (CE-based, including the non-monotonic temperature-CE profile as the unique experimental signature of the arctic optimum). The non-monotonic profile — LT CE increasing as temperature falls from $+25^\circ\text{C}$ to -27°C , then decreasing — is the prediction that no prior framework can generate. Its confirmation would constitute the first experimental demonstration of a deliberately engineered temperature-responsive SCE gap in a Li metal battery electrolyte.

The full derivation record, MU session confirmation, and all pre-registered conditions are archived at:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>

Data and derivation record availability. All pre-registered predictions, falsification conditions, temperature coefficient derivations, SCE ceiling calculation, MU session Query 4 and Query 7 records, and population estimates are archived with full timestamped commit history at:

<https://github.com/Eric-Robert-Lawson/attractor-oncology>

The repository commit history constitutes the priority and pre-registration record for all claims in this paper.

Relationship to other papers in this series. This is Preprint 5 of the SCE Framework series. Preprints 1–4 are available at the same repository. All papers are self-contained. The geometric structure connecting all papers, mechanisms, candidates, and derivations is navigable from the repository index.

Competing interests. The author declares no competing financial interests.

Author contributions. E.R.L.: conceptualisation, derivation, temperature coefficient analysis, SCE ceiling derivation, writing.

A SCE Ceiling Derivation Detail

The practical SCE ceiling of 1.75 is derived by computing the maximum SCE achievable when the dominant configuration fraction is constrained to $p_{\max} \geq 0.20$.

For a system with n configurations and dominant fraction $p_{\max} = 0.20$, the maximum entropy is achieved when all remaining configurations share equal weight: $p_i = (1 - 0.20)/(n - 1) = 0.80/(n - 1)$ for $i \neq 1$.

$$\text{SCE}_{\max}(n, p_{\max} = 0.20) = -0.20 \ln(0.20) - (n - 1) \cdot \frac{0.80}{n - 1} \ln\left(\frac{0.80}{n - 1}\right) \quad (25)$$

For $n = 5$ (a three-component ether system with 4 SSIP sub-types and 1 CIP type):

$$\text{SCE}_{\max}(5, 0.20) = -0.20 \ln(0.20) - 4 \times 0.20 \times \ln(0.20) \quad (26)$$

$$= -5 \times 0.20 \times \ln(0.20) \quad (27)$$

$$= -\ln(0.20) = \ln(5) = 1.609 \quad (28)$$

For $n = 7$ (including AGG configurations):

$$\text{SCE}_{\max}(7, 0.20) = -0.20 \ln(0.20) - 6 \times \frac{0.80}{6} \ln\left(\frac{0.80}{6}\right) = 0.322 + 1.389 = 1.711 \quad (29)$$

Allowing $p_{\max}$ to be as low as 0.15 in highly mixed three-component systems:

$$\text{SCE}_{\max}(7, 0.15) \approx 1.80 \quad (30)$$

The practical ceiling of 1.75 represents the midpoint of the range $[1.711, 1.80]$, appropriate for the three-component DME/TMP/TMB system where $n \approx 6-7$ and $p_{\max} \geq 0.17$ is physically constrained by the DME bidentate enthalpic preference.

Table 3: Physical and electrochemical properties of components in Arctic Candidate A.

Component	Role	mp (°C)	bp (°C)	ρ (g mL ⁻¹)	SCE (1M LiFSI)
DME	Base solvent	-58	85	0.867	1.24
TMP	ESP co-solvent	-46	197	1.214	—
TMB	Anion receptor	-29	68	0.915	—
LiFSI	Salt	—	—	—	—

B Solvent Properties: Three-Component System

All properties from [2]. The mixture melting point of DME/TMP/TMB 3:1:0.5 is below -50°C (estimated from ideal mixing of pure-component melting points), confirming liquid-phase operation at -30°C .

C Pre-Registered Falsification Conditions (Verbatim)

Arctic Candidate A (LiFSI 1.2 M, DME/TMP/TMB 3:1:0.5 v/v/v):

Structural conditions (Raman):

- S1: RT SCE in [1.51, 1.65].
- S2: SCE at -30°C in [1.43, 1.50].
- S3: SCE decreases monotonically from $+25$ to -30°C .
- S4: FSI⁻ Raman symmetric stretch shifts toward SSIP as temperature decreases.

Performance conditions (CE):

- P1: CE at $-30^{\circ}\text{C} > \text{CE at } -20^{\circ}\text{C}$.
- P2: CE at $-30^{\circ}\text{C} \geq 60\%$.
- P3: CE at $+25^{\circ}\text{C} < 73\%$.
- P4: CE_{LT} shows a maximum between -20 and -35°C in the temperature scan.

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